

Roshan Taneja

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EDUCATION

UC Berkeley

2025 – 2028

Bachelor of Science in Electrical Engineering & Computer Science (EECS)

Berkeley, CA

Courses: Structure & Interpretation of Computer Programs, Data Structures & Algorithms, Linear Algebra, Discrete Mathematics & Probability Theory, Multivariable Calculus, Introduction to Circuits & Devices, Foundations of Signals, Dynamical Systems, & Information Processing, Introduction to Amateur Rocketry

Harvard University – Spring 2025

Spatial Research Certificate

UCLA CSSI – Summer 2024

Intermediate Data Science

MIT Beaver Works – Spring 2024

Remote Sensing

Sacred Heart Preparatory

2021 – 2025

High School Diploma — GPA 3.98 · ACT 36

Atherton, CA

EXPERIENCE & PROJECTS

Backend Engineer

2026 – Present

Statewide Database

Berkeley, CA

- Built end-to-end voter-registration ingestion pipeline in **Dagster** + **DuckDB**
- processing fixture files through validation (Pydantic) and loading with correct category/geography detection.
- Reduced manual validation overhead by automating schema checks across multi-format registration files.

Avionics Engineer

2025 – Present

Cal Space Technologies and Rocketry (STAR)

Berkeley, CA

- Designing avionics systems for high-power rockets (~30,000 ft apogee) with custom PCBs via Altium Designer
- Soldering and testing high-/low-voltage boards under strict mass and volume constraints; firmware in ESP32/C++

Research Intern — Center for Geographic Analysis

2024 – 2025

Harvard University

Cambridge, MA

- with Prof. Wang (CGA), **satellite ML models** identifying Maasai settlements across **10,000+ image tiles**
- Optimizing placement of rainwater harvesting units in Tanzania. Expanded model to new geographies; work underpins active humanitarian deployment.

Data Science Researcher

2024 – 2025

Cost of Colonialism (COCO)

Remote

- Built **OCR + NLP semantic search** over 1,600s–1,800s Dutch East India Company shipping manifests; **embedding-based retrieval** with historian-facing query UI.
- Automated entity extraction across noisy archival scans, reducing manual review time from weeks to minutes.

Youth President, Water Programs

2021 – 2025

Maji Wells

Losimingori, TZ

- Led development of an **ML satellite image classifier** to locate Maasai communities for infrastructure planning.
- **Fundraised \$300,000** to deploy **500 rainwater harvesting solutions** directly serving **30,000 Maasai** across Tanzania
- Core work behind the 2024 United Nations SDG #6 Award and Children's Peace Prize nomination.

PUBLICATIONS, PATENTS & CONFERENCES

AICC @ NeurIPS | *Poster Presentation*

2025

- Copenhagen, Denmark

NeurIPS 2024 | *"Machine Learning for Social Impact" (1 of 4 Winners)*

2024

- 14,000 attendees; selected from global submissions across all career stages.

Provisional Patent | *Object Detection in Satellite Imagery for Optimising Rainwater Unit Placement*

2024

- US Patent Application 63/703,232 · Published in NHSJS

ML4EO 2024 | *University of Exeter, UK – Sole High-school Keynote Presenter*

2024

Publication | *Impact of Rainwater Harvesting on 4,500+ Maasai (NHSJS)*

2024

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, AppScript, SQL(Postgres)

Geospatial: ArcGIS Pro, Google Earth Engine, PostGIS, GeoPandas, AlphaEarth, MCDA, SOAR, mAP/F1 evaluation

Infrastructure: Git, Docker, Linux, DuckDB, Dagster, Pydantic, Full-Stack Web Development

Hardware: Altium Designer, KiCad, ESP32, PCB Design, CAD/CAM (Fusion 360, SolidWorks), CNC, 3D Printing, Laser Cutting

Referenes available upon request.